

EE 2101: Quiz 3

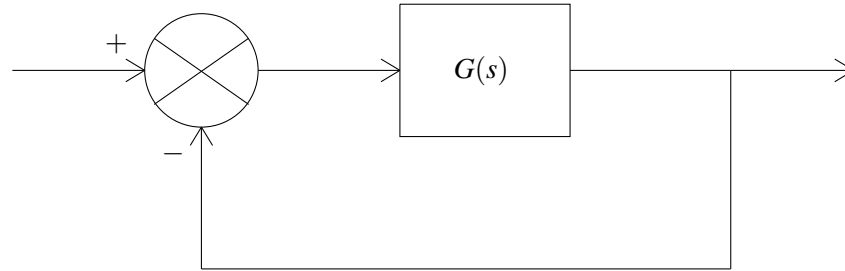


Figure 1: Control System

1. Consider the system with the transfer function

[Marks: 3]

$$T(s) = \frac{50(s+1)}{(s+5)(s^2+s+7)}$$

A sinusoidal signal $5\sin(10t + 70^\circ)$ is applied to this system. Find the steady-state output of the system.

2. Consider a control system with transfer function

[Marks: 3]

$$T(s) = \frac{16}{s^2 + 4.8s + 16}$$

An input signal $2\sin(\omega t)$ is applied to this system. Find the value of ω for which the system provides maximum amplitude amplification of the input signal. Also find the corresponding amplification factor.

3. A control system with the following transfer function was purchased a few years ago

[Marks: 5]

$$T(s) = \frac{(s+5)}{s(s+1)(s^2+4s+49)}$$

Now, the system has developed a delay of 2 seconds between the application of input signal and the change in output response. Let $T_1(s)$ denote the transfer function of the degraded system. Then,

- Draw the asymptotic Bode plot of $T_1(s)$.
- Based on this plot, compute the magnitude (dB) and the phase angle (degrees) of $T_1(s)$ for $\omega = 2$ rad/s and $\omega = 10$ rad/s. Make a table of these values.

4. Consider the control system shown in Figure 1 with

[Marks: 4]

$$G(s) = \frac{s+2}{(s+1)(s^2+4s+5)(s^2+8s+25)}$$

Find the gain margin and the phase crossover frequency for this system. Note that (approximate) computation based on asymptotic Bode plot will get zero marks.

5. Consider the control system shown in Figure 1 with

[Marks: 5]

$$G(s) = \frac{K(s^4 + 15s^3 + 74s^2 + 132s + 72)}{s^4 + 9s^3 + 3s^2 - 145s - 300}$$

(a) Draw the Nyquist plot for this system.

(b) By using above Nyquist plot, find the non-negative range of K for which this system is BIBO stable.

All the Best!